

Business Problems:

Problem 1: How do daily trip volumes compare between January 2024 and January 2025?

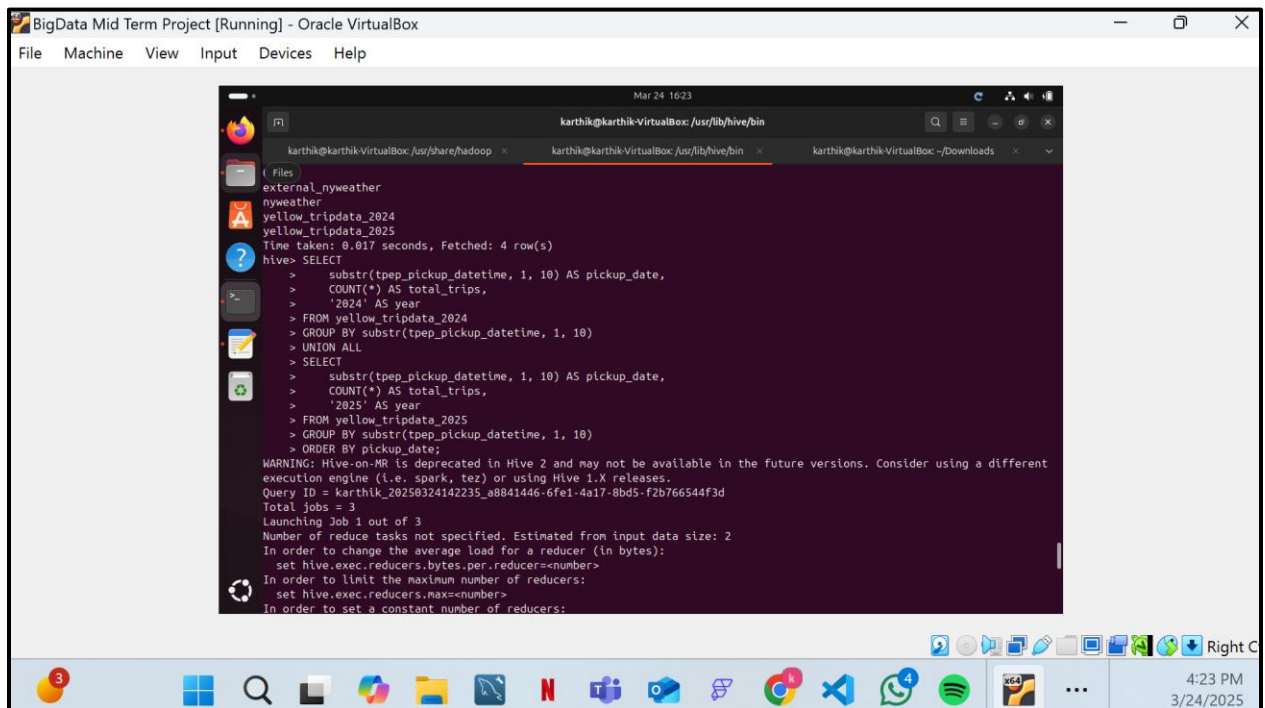
Problem 2: Which pickup locations (PULocationID) have the highest trip counts, and do these change from 2024 to 2025?

Problem 3: How do average fares per day vary between January 2024 and January 2025?

Problem 4: (Optional) What are the average trip speeds, and how do they reflect traffic or operational efficiency differences between the two periods?

Karthik Queries

Query 1

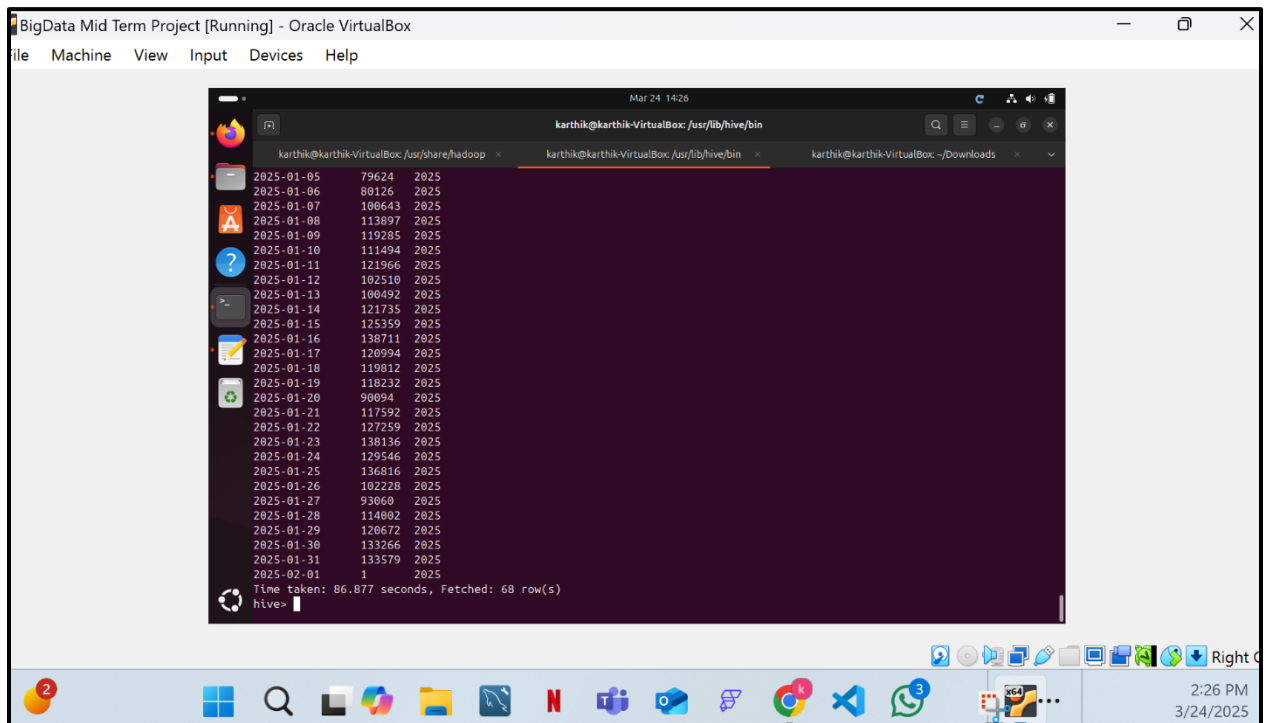
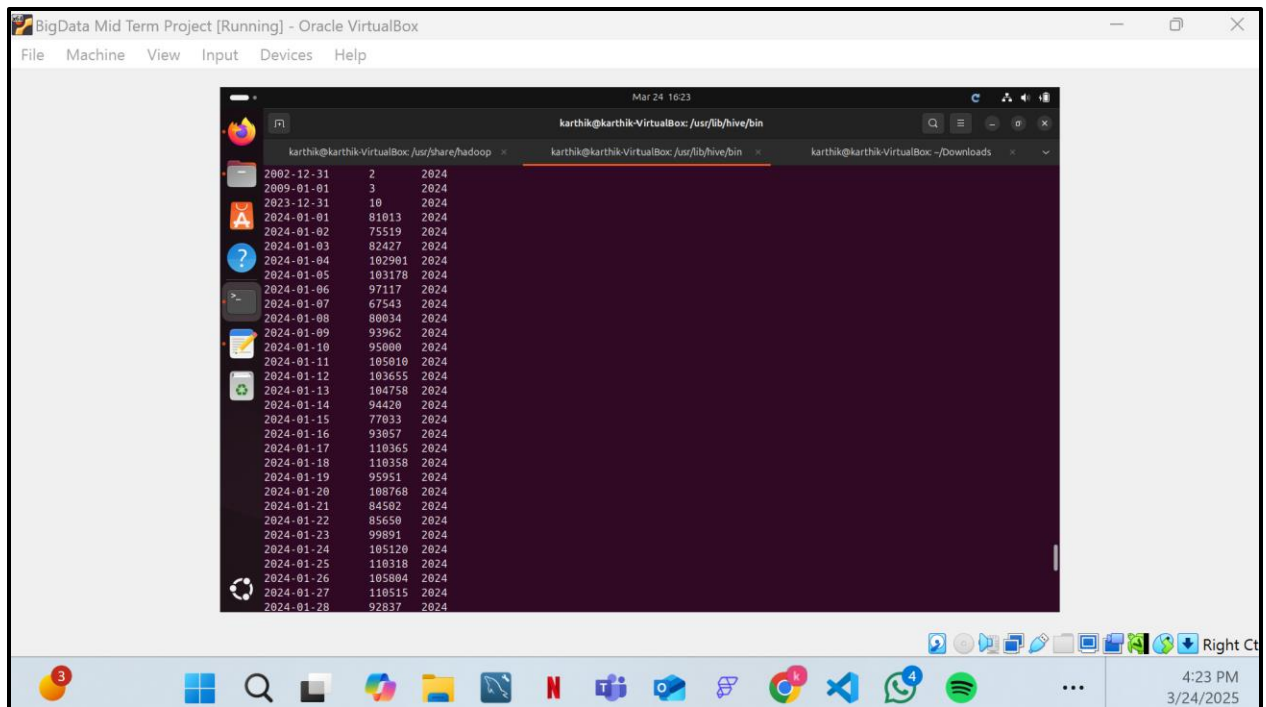


The screenshot shows a Windows desktop with a VirtualBox window titled "BigData Mid Term Project [Running] - Oracle VirtualBox". Inside the VirtualBox, there is a terminal window with a dark background. The terminal shows the execution of a Hive query. The query is a SELECT statement that joins data from two tables, yellow_tripdata_2024 and yellow_tripdata_2025, based on pickup date and year. The query is executed using the Hive command line interface. The output shows the query ID, total jobs, and the launch of Job 1 out of 3. The terminal also displays a warning about Hive-on-MR being deprecated and suggestions for alternative execution engines or Hive versions. The bottom of the screenshot shows the Windows taskbar with various application icons and the system clock indicating 4:23 PM on 3/24/2025.

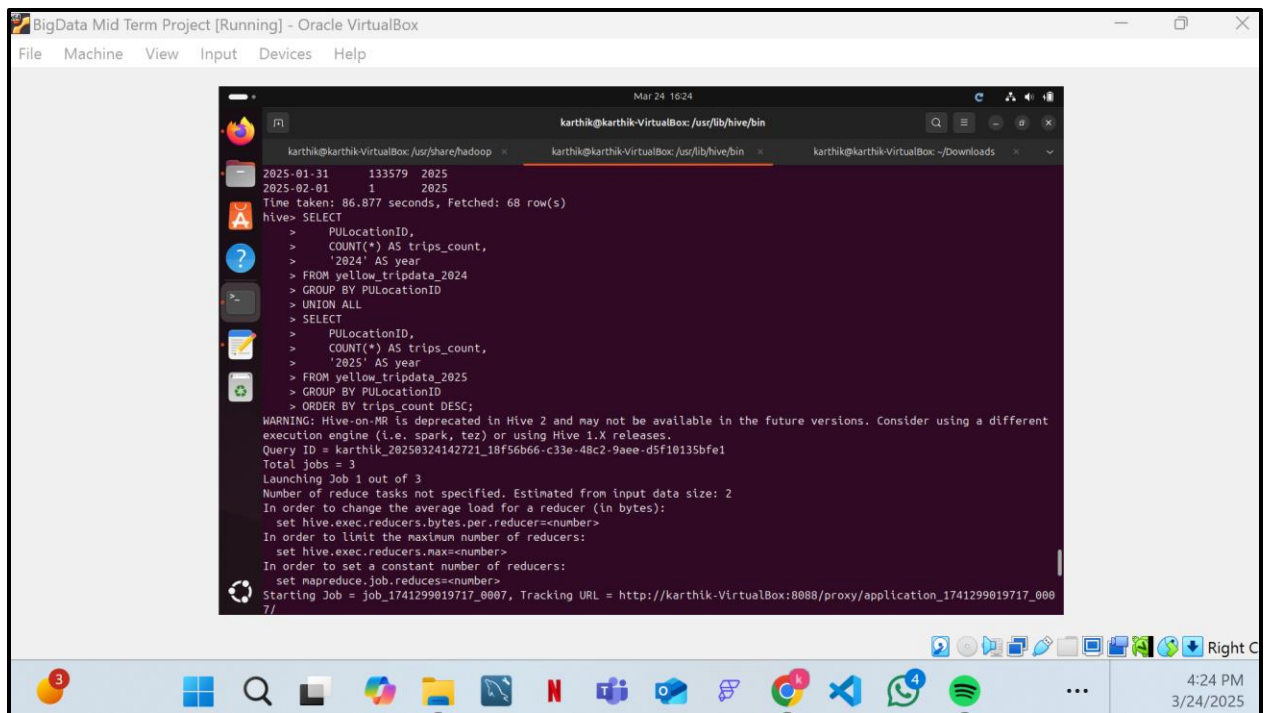
```
karthik@karthik-VirtualBox: /usr/lib/hive/bin
karthik@karthik-VirtualBox: /usr/share/hadoop
karthik@karthik-VirtualBox: /usr/lib/hive/bin
karthik@karthik-VirtualBox: ~Downloads

Files
external_nyweather
nyweather
yellow_tripdata_2024
yellow_tripdata_2025
Time taken: 0.017 seconds, Fetched: 4 row(s)
hive> SELECT
>   substr(tppep_pickup_datetime, 1, 10) AS pickup_date,
>   COUNT(*) AS total_trips,
>   '2024' AS year
> FROM yellow_tripdata_2024
> GROUP BY substr(tppep_pickup_datetime, 1, 10)
> UNION ALL
> SELECT
>   substr(tppep_pickup_datetime, 1, 10) AS pickup_date,
>   COUNT(*) AS total_trips,
>   '2025' AS year
> FROM yellow_tripdata_2025
> GROUP BY substr(tppep_pickup_datetime, 1, 10)
> ORDER BY pickup_date;

WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different
execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = karthik_20250324142235_a8841446-6fe1-4a17-8bd5-f2b766544f3d
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks not specified. Estimated from input data size: 2
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
```



Query 2

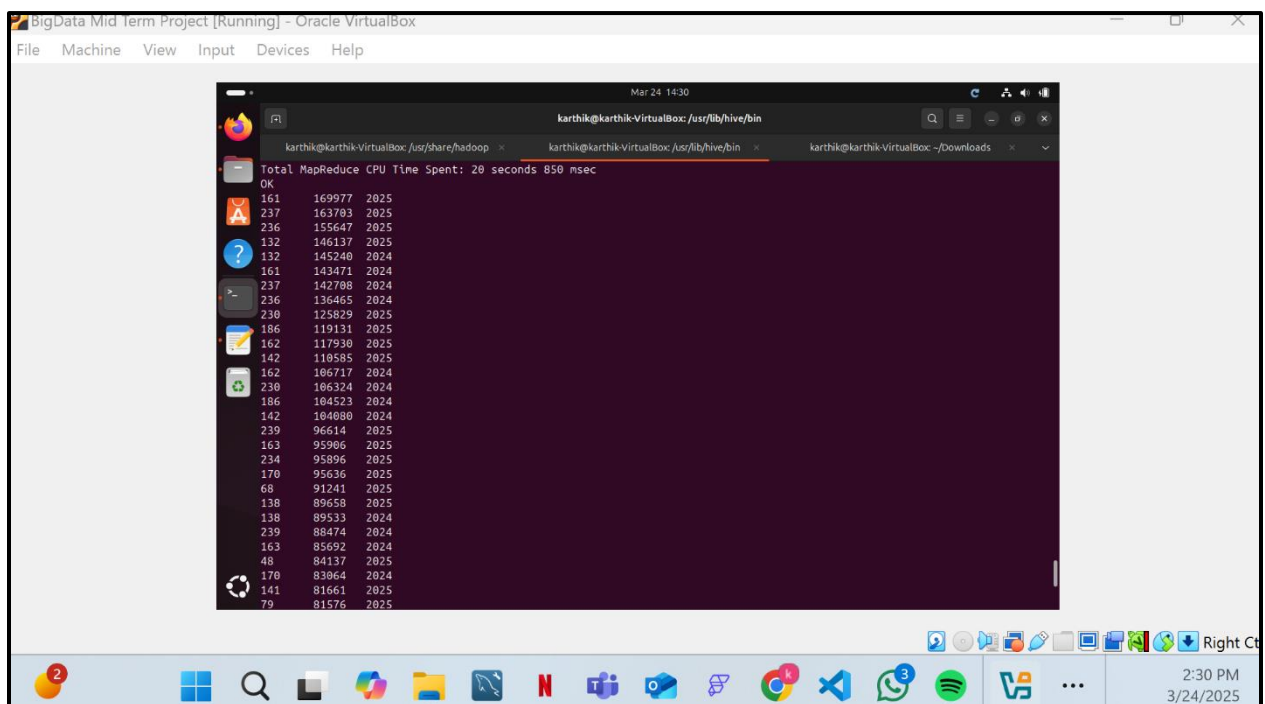


```
BigData Mid Term Project [Running] - Oracle VirtualBox
File Machine View Input Devices Help

karthik@karthik-VirtualBox: /usr/lib/hive/bin

2025-01-31 133579 2025
2025-02-01 1 2025
Time taken: 86.877 seconds, Fetched: 68 row(s)
hive> SELECT
>   PULocationID,
>   COUNT(*) AS trips_count,
>   '2024' AS year
> FROM yellow_tripdata_2024
> GROUP BY PULocationID
> UNION ALL
> SELECT
>   PULocationID,
>   COUNT(*) AS trips_count,
>   '2025' AS year
> FROM yellow_tripdata_2025
> GROUP BY PULocationID
> ORDER BY trips_count DESC;

WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different
execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = karthik_20250324142721_18f56b66-c33e-48c2-9aee-d5f10135bf1
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks not specified. Estimated from input data size: 2
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1741299019717_0007, Tracking URL = http://karthik-VirtualBox:8088/proxy/application_1741299019717_0007/
```

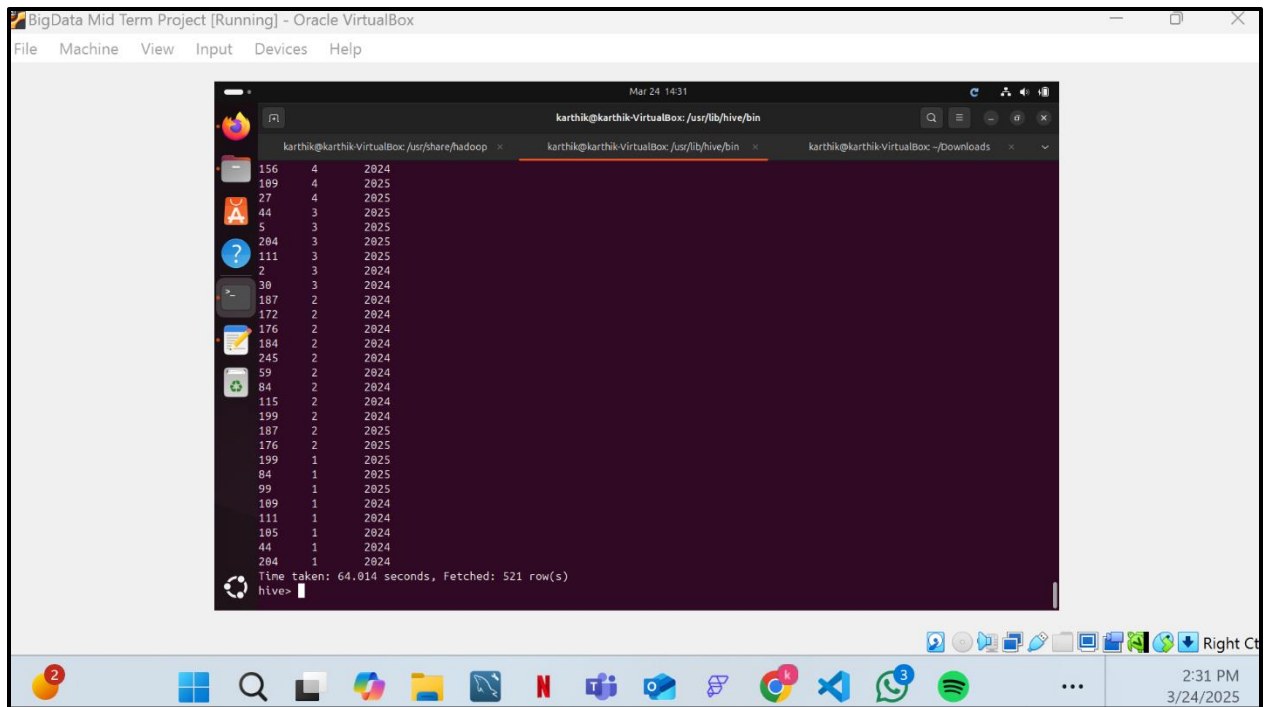


```
BigData Mid Term Project [Running] - Oracle VirtualBox
File Machine View Input Devices Help

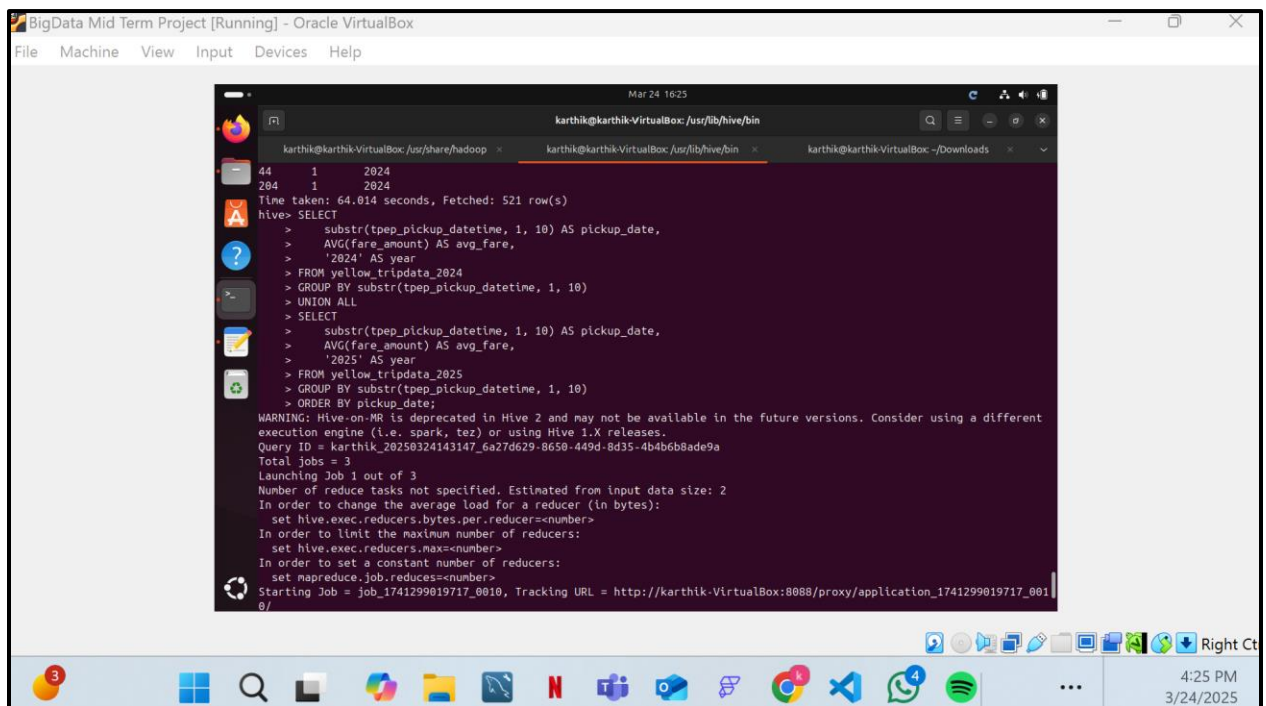
karthik@karthik-VirtualBox: /usr/lib/hive/bin

karthik@karthik-VirtualBox: /usr/share/hadoop > karthik@karthik-VirtualBox: /usr/lib/hive/bin > karthik@karthik-VirtualBox: ~Downloads >

Total MapReduce CPU Time Spent: 28 seconds 850 msec
OK
161 169977 2025
237 163783 2025
236 155647 2025
132 146137 2025
132 145240 2024
161 143471 2024
237 142788 2024
236 136465 2024
230 125829 2025
186 119131 2025
162 117930 2025
142 110585 2025
162 106717 2024
230 106324 2024
186 104523 2024
142 104080 2024
239 96614 2025
163 95906 2025
234 95896 2025
170 95636 2025
68 91241 2025
138 89658 2025
138 89533 2024
239 88474 2024
163 85692 2024
48 84137 2025
170 83964 2024
141 81661 2025
79 81576 2025
```



Query 3



BigData Mid Term Project [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 24 14:36

karthik@karthik-VirtualBox: /usr/lib/hive/bin

karthik@karthik-VirtualBox: /usr/share/hadoop

karthik@karthik-VirtualBox: ~/Downloads

Total MapReduce CPU Time Spent: 34 seconds 528 msec

OK

2002-12-31	0.0	2024	2024
2009-01-01	33.33333333333336	2024	2024
2023-12-31	14.41	2024	2024
2024-01-01	21.788021058348747	2024	2024
2024-01-02	20.967073451714302	2024	2024
2024-01-03	19.66402258968544	2024	2024
2024-01-04	18.421313495495873	2024	2024
2024-01-05	17.83245468995311	2024	2024
2024-01-06	17.310473449550734	2024	2024
2024-01-07	19.45297854699959	2024	2024
2024-01-08	18.698506009945955	2024	2024
2024-01-09	16.89926153125751	2024	2024
2024-01-10	17.920641684210477	2024	2024
2024-01-11	18.55664193806312	2024	2024
2024-01-12	18.548339973952157	2024	2024
2024-01-13	17.276198858321262	2024	2024
2024-01-14	17.870277165854585	2024	2024
2024-01-15	19.298005789726666	2024	2024
2024-01-16	19.172022523829376	2024	2024
2024-01-17	18.203946359806263	2024	2024
2024-01-18	17.895973558781506	2024	2024
2024-01-19	17.2211299517463	2024	2024
2024-01-20	16.53430209252731	2024	2024
2024-01-21	18.43072038517063	2024	2024
2024-01-22	18.535767474022843	2024	2024
2024-01-23	17.677200548597973	2024	2024
2024-01-24	17.71617958913245	2024	2024
2024-01-25	18.263443046465962	2024	2024
2024-01-26	17.815411326603893	2024	2024

2:36 PM
3/24/2025

BigData Mid Term Project [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 24 14:37

karthik@karthik-VirtualBox: /usr/lib/hive/bin

karthik@karthik-VirtualBox: /usr/share/hadoop

karthik@karthik-VirtualBox: ~/Downloads

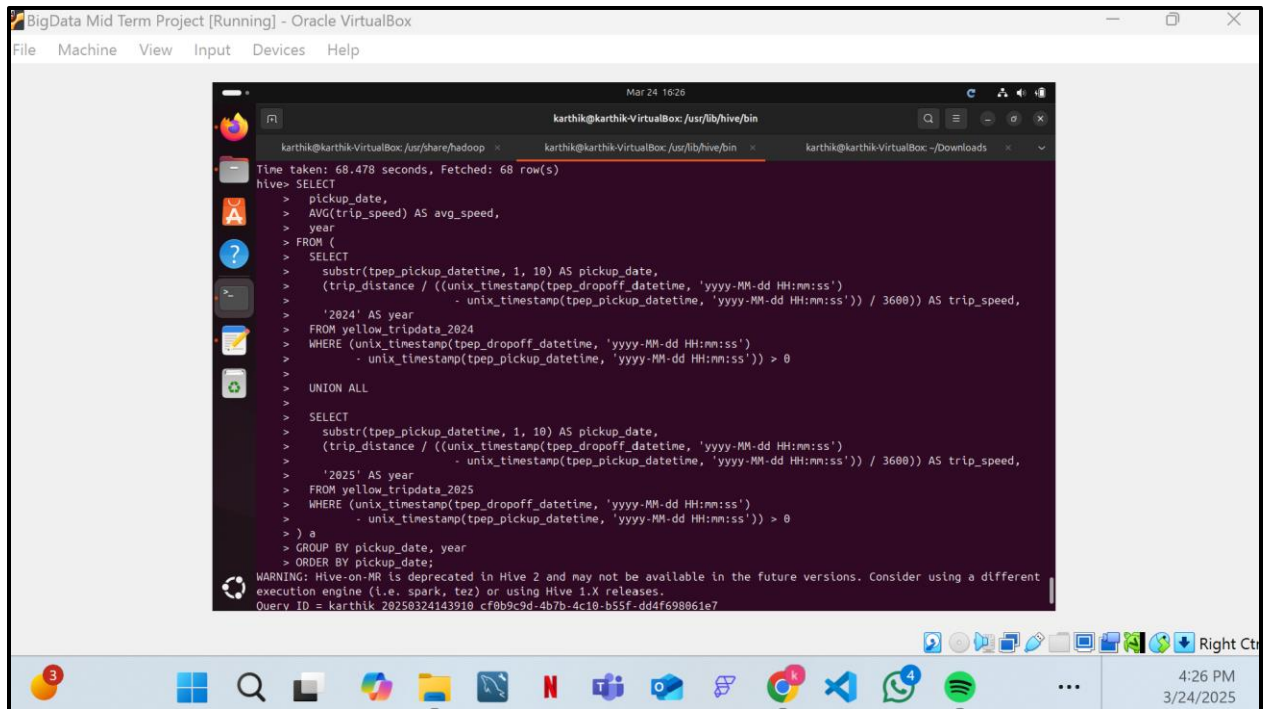
2025-01-05	19.053785667638053	2025	2025
2025-01-06	18.32262137134027	2025	2025
2025-01-07	16.752821159941846	2025	2025
2025-01-08	16.221754743320844	2025	2025
2025-01-09	16.598833298403136	2025	2025
2025-01-10	16.9549432256449	2025	2025
2025-01-11	15.857532345079944	2025	2025
2025-01-12	17.31504243488442	2025	2025
2025-01-13	17.084066194324112	2025	2025
2025-01-14	16.428041812133113	2025	2025
2025-01-15	16.605769350425483	2025	2025
2025-01-16	17.1370135028946	2025	2025
2025-01-17	16.909112435327364	2025	2025
2025-01-18	15.636025523319965	2025	2025
2025-01-19	15.64282613043966	2025	2025
2025-01-20	27.96109774235824	2025	2025
2025-01-21	16.71816968841415	2025	2025
2025-01-22	16.053786372673088	2025	2025
2025-01-23	16.368340258875442	2025	2025
2025-01-24	16.407236194093215	2025	2025
2025-01-25	15.576826686937268	2025	2025
2025-01-26	17.34383172907633	2025	2025
2025-01-27	17.345586395873777	2025	2025
2025-01-28	16.341610936650344	2025	2025
2025-01-29	16.410052157916016	2025	2025
2025-01-30	16.83383263548075	2025	2025
2025-01-31	16.4420227730409	2025	2025
2025-02-01	6.5	2025	2025

Time taken: 68.478 seconds, Fetched: 68 row(s)

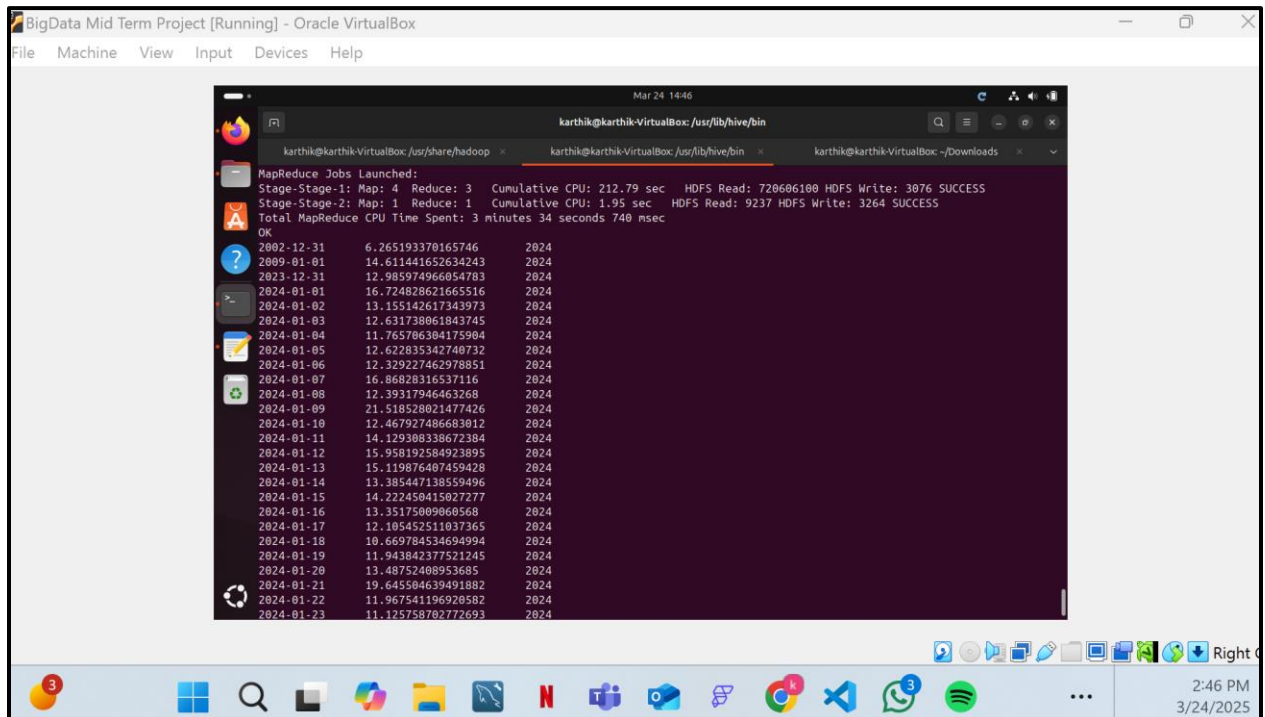
hive>

2:37 PM
3/24/2025

Query 4



```
Time taken: 68.478 seconds, Fetched: 68 row(s)
hive> SELECT
>   pickup_date,
>   AVG(trip_speed) AS avg_speed,
>   year
> FROM (
>   SELECT
>     substr(tpcp_pickup_datetime, 1, 10) AS pickup_date,
>     (trip_distance / ((unix_timestamp(tpcp_dropoff_datetime, 'yyyy-MM-dd HH:mm:ss')
>       - unix_timestamp(tpcp_pickup_datetime, 'yyyy-MM-dd HH:mm:ss')) / 3600)) AS trip_speed,
>     '2024' AS year
>   FROM yellow_tripdata_2024
>   WHERE (unix_timestamp(tpcp_dropoff_datetime, 'yyyy-MM-dd HH:mm:ss')
>     - unix_timestamp(tpcp_pickup_datetime, 'yyyy-MM-dd HH:mm:ss')) > 0
>   UNION ALL
>   SELECT
>     substr(tpcp_pickup_datetime, 1, 10) AS pickup_date,
>     (trip_distance / ((unix_timestamp(tpcp_dropoff_datetime, 'yyyy-MM-dd HH:mm:ss')
>       - unix_timestamp(tpcp_pickup_datetime, 'yyyy-MM-dd HH:mm:ss')) / 3600)) AS trip_speed,
>     '2025' AS year
>   FROM yellow_tripdata_2025
>   WHERE (unix_timestamp(tpcp_dropoff_datetime, 'yyyy-MM-dd HH:mm:ss')
>     - unix_timestamp(tpcp_pickup_datetime, 'yyyy-MM-dd HH:mm:ss')) > 0
> ) a
> GROUP BY pickup_date, year
> ORDER BY pickup_date;
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different
execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = karthik_20250324143910_cf0b9c9d-4b7b-4c10-b55f-dd4f698061e7
```



```
MapReduce Jobs Launched:
Stage-Stage-1: Map: 4 Reduce: 3 Cumulative CPU: 212.79 sec HDFS Read: 720606100 HDFS Write: 3076 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 1.95 sec HDFS Read: 9237 HDFS Write: 3264 SUCCESS
Total MapReduce CPU Time Spent: 3 minutes 34 seconds 740 msec
OK
2002-12-31 6.265193370165746 2024
2009-01-01 14.611441652634243 2024
2023-12-31 12.985974966054783 2024
2024-01-01 16.724828621665516 2024
2024-01-02 13.155142617343973 2024
2024-01-03 12.631738061843745 2024
2024-01-04 11.765706304175904 2024
2024-01-05 12.622835342740732 2024
2024-01-06 12.329227462978851 2024
2024-01-07 16.86828316537116 2024
2024-01-08 12.39317946463268 2024
2024-01-09 21.518528021477426 2024
2024-01-10 12.467927486683812 2024
2024-01-11 14.129308338672384 2024
2024-01-12 15.958192584923895 2024
2024-01-13 15.119876407459428 2024
2024-01-14 13.385447138559496 2024
2024-01-15 14.222450415027277 2024
2024-01-16 13.35175009060568 2024
2024-01-17 12.105452511037365 2024
2024-01-18 10.669784534694994 2024
2024-01-19 11.943842377521245 2024
2024-01-20 13.48752408953685 2024
2024-01-21 19.645504639491882 2024
2024-01-22 11.967541196920582 2024
2024-01-23 11.125758702772693 2024
```

BigData Mid Term Project [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 24 14:47

karthik@karthik-VirtualBox: /usr/lib/hive/bin

karthik@karthik-VirtualBox: /usr/share/hadoop

karthik@karthik-VirtualBox: /usr/lib/hive/bin

karthik@karthik-VirtualBox: ~Downloads

2025-01-05	14.084249558805677	2025
2025-01-06	14.65725412496656	2025
2025-01-07	12.446657376952844	2025
2025-01-08	10.937688418036151	2025
2025-01-09	33.99729022691588	2025
2025-01-10	18.723461238692952	2025
2025-01-11	29.406345093961404	2025
2025-01-12	38.06265415853462	2025
2025-01-13	40.544944932646594	2025
2025-01-14	49.55014219903404	2025
2025-01-15	27.102219879099454	2025
2025-01-16	36.72720392280287	2025
2025-01-17	28.85901586259742	2025
2025-01-18	23.331034979105773	2025
2025-01-19	37.27286914489086	2025
2025-01-20	17.098110266246945	2025
2025-01-21	25.680994497803727	2025
2025-01-22	44.70842822610222	2025
2025-01-23	24.838290754236922	2025
2025-01-24	26.428817292906146	2025
2025-01-25	18.9906184037476	2025
2025-01-26	28.330707887261426	2025
2025-01-27	15.71013716214696	2025
2025-01-28	10.562538192836273	2025
2025-01-29	16.57806090453346	2025
2025-01-30	24.004695825969174	2025
2025-01-31	27.753154016815657	2025
2025-02-01	13.371428571420572	2025

Time taken: 92.755 seconds, Fetched: 68 row(s)

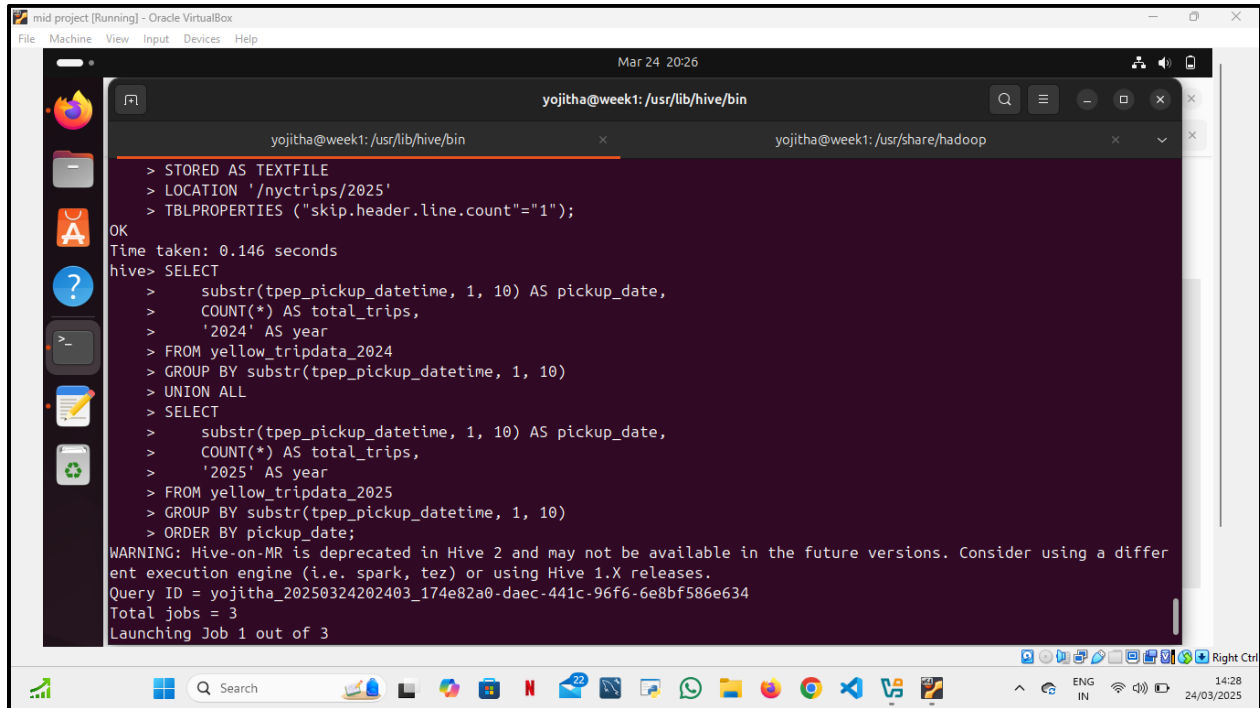
hive>

Right Click

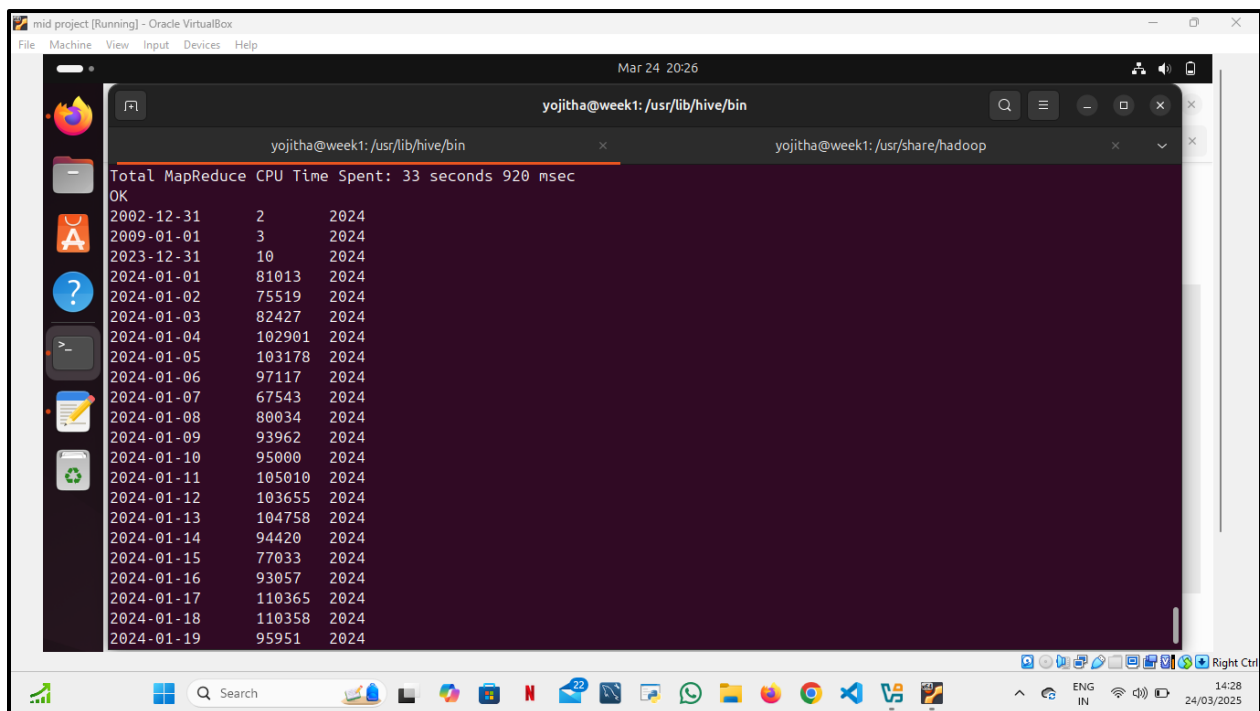
2:47 PM
3/24/2025

Yojitha Queries

Query 1



```
yojitha@week1: /usr/lib/hive/bin
> STORED AS TEXTFILE
> LOCATION '/nyc trips/2025'
> TBLPROPERTIES ("skip.header.line.count"="1");
OK
Time taken: 0.146 seconds
hive> SELECT
>   substr(tpep_pickup_datetime, 1, 10) AS pickup_date,
>   COUNT(*) AS total_trips,
>   '2024' AS year
> FROM yellow_tripdata_2024
> GROUP BY substr(tpep_pickup_datetime, 1, 10)
> UNION ALL
> SELECT
>   substr(tpep_pickup_datetime, 1, 10) AS pickup_date,
>   COUNT(*) AS total_trips,
>   '2025' AS year
> FROM yellow_tripdata_2025
> GROUP BY substr(tpep_pickup_datetime, 1, 10)
> ORDER BY pickup_date;
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = yojitha_20250324202403_174e82a0-daec-441c-96f6-6e8bf586e634
Total jobs = 3
Launching Job 1 out of 3
```



```
Total MapReduce CPU Time Spent: 33 seconds 920 msec
OK
2002-12-31      2      2024
2009-01-01      3      2024
2023-12-31     10      2024
2024-01-01    81013      2024
2024-01-02    75519      2024
2024-01-03    82427      2024
2024-01-04   102901      2024
2024-01-05   103178      2024
2024-01-06    97117      2024
2024-01-07    67543      2024
2024-01-08    80034      2024
2024-01-09    93962      2024
2024-01-10    95000      2024
2024-01-11   105010      2024
2024-01-12   103655      2024
2024-01-13   104758      2024
2024-01-14    94420      2024
2024-01-15    77033      2024
2024-01-16    93057      2024
2024-01-17   110365      2024
2024-01-18   110358      2024
2024-01-19    95951      2024
```



```
mid project [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 24 20:26

yojitha@week1: /usr/lib/hive/bin
yojitha@week1: /usr/lib/hive/bin
yojitha@week1: /usr/share/hadoop

2025-01-11 121966 2025
2025-01-12 102510 2025
2025-01-13 100492 2025
2025-01-14 121735 2025
2025-01-15 125359 2025
2025-01-16 138711 2025
2025-01-17 120994 2025
2025-01-18 119812 2025
2025-01-19 118232 2025
2025-01-20 90094 2025
2025-01-21 117592 2025
2025-01-22 127259 2025
2025-01-23 138136 2025
2025-01-24 129546 2025
2025-01-25 136816 2025
2025-01-26 102228 2025
2025-01-27 93060 2025
2025-01-28 114002 2025
2025-01-29 120672 2025
2025-01-30 133266 2025
2025-01-31 133579 2025
2025-02-01 1 2025
Time taken: 100.669 seconds, Fetched: 68 row(s)
hive>
```

Query 2

```
mid project [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 24 20:28

yojitha@week1: /usr/lib/hive/bin
yojitha@week1: /usr/lib/hive/bin
yojitha@week1: /usr/share/hadoop

2025-02-01 1 2025
Time taken: 100.669 seconds, Fetched: 68 row(s)
hive> SELECT
>   PULocationID,
>   COUNT(*) AS trips_count,
>   '2024' AS year
> FROM yellow_tripdata_2024
> GROUP BY PULocationID
> UNION ALL
> SELECT
>   PULocationID,
>   COUNT(*) AS trips_count,
>   '2025' AS year
> FROM yellow_tripdata_2025
> GROUP BY PULocationID
> ORDER BY trips_count DESC;
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = yojitha_20250324202641_45c5ea63-a337-4bf3-bb83-8ac74bc22450
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks not specified. Estimated from input data size: 2
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
```

mid project [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 24 20:28

yojitha@week1: /usr/lib/hive/bin

yojitha@week1: /usr/lib/hive/bin

yojitha@week1: /usr/share/hadoop

Total MapReduce CPU Time Spent: 27 seconds 550 msec

OK

161	169977	2025
237	163703	2025
236	155647	2025
132	146137	2025
132	145240	2024
161	143471	2024
237	142708	2024
236	136465	2024
230	125829	2025
186	119131	2025
162	117930	2025
142	110585	2025
162	106717	2024
230	106324	2024
186	104523	2024
142	104080	2024
239	96614	2025
163	95906	2025
234	95896	2025
170	95636	2025
68	91241	2025
138	89658	2025

14:30 24/03/2025

mid project [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 24 20:28

yojitha@week1: /usr/lib/hive/bin

yojitha@week1: /usr/lib/hive/bin

yojitha@week1: /usr/share/hadoop

111	3	2025
2	3	2024
30	3	2024
187	2	2024
172	2	2024
176	2	2024
184	2	2024
245	2	2024
59	2	2024
84	2	2024
115	2	2024
199	2	2024
187	2	2025
176	2	2025
199	1	2025
84	1	2025
99	1	2025
109	1	2024
111	1	2024
105	1	2024
44	1	2024
204	1	2024

Time taken: 94.818 seconds, Fetched: 521 row(s)

hive>

14:30 24/03/2025

Query 3

```
mid project [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 24 20:31

yojitha@week1: /usr/lib/hive/bin

yojitha@week1: /usr/lib/hive/bin
yojitha@week1: /usr/share/hadoop

hive> SELECT
>   substr(tpep_pickup_datetime, 1, 10) AS pickup_date,
>   AVG(fare_amount) AS avg_fare,
>   '2024' AS year
> FROM yellow_tripdata_2024
> GROUP BY substr(tpep_pickup_datetime, 1, 10)
> UNION ALL
> SELECT
>   substr(tpep_pickup_datetime, 1, 10) AS pickup_date,
>   AVG(fare_amount) AS avg_fare,
>   '2025' AS year
> FROM yellow_tripdata_2025
> GROUP BY substr(tpep_pickup_datetime, 1, 10)
> ORDER BY pickup_date;

WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = yojitha_20250324202905_a43cd383-726d-4277-affa-52d93487f5b8
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks not specified. Estimated from input data size: 2
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
```

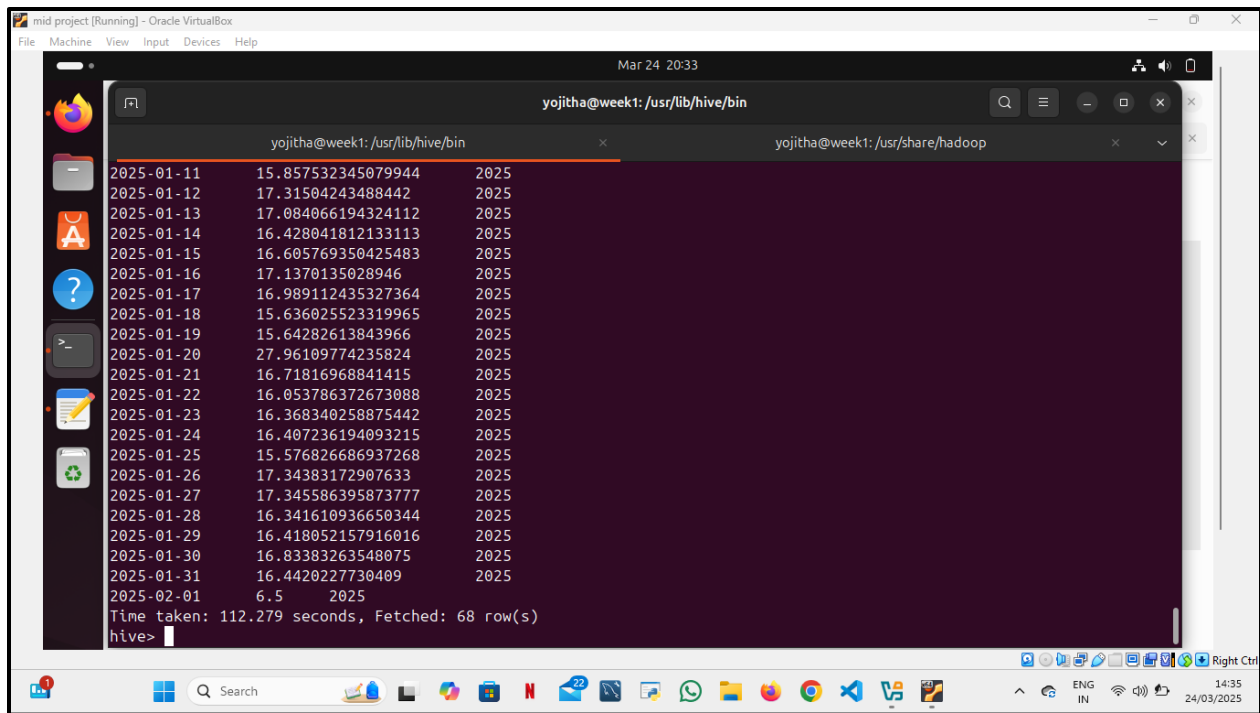
```
mid project [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 24 20:33

yojitha@week1: /usr/lib/hive/bin

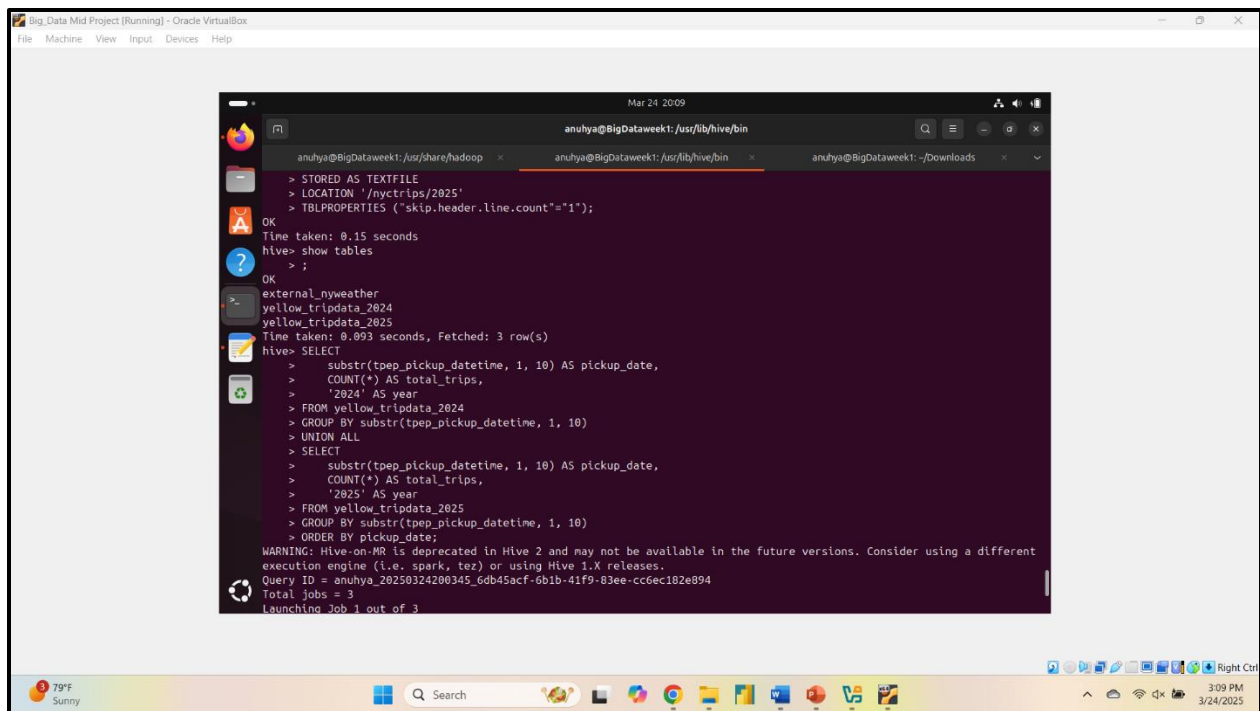
yojitha@week1: /usr/lib/hive/bin
yojitha@week1: /usr/share/hadoop

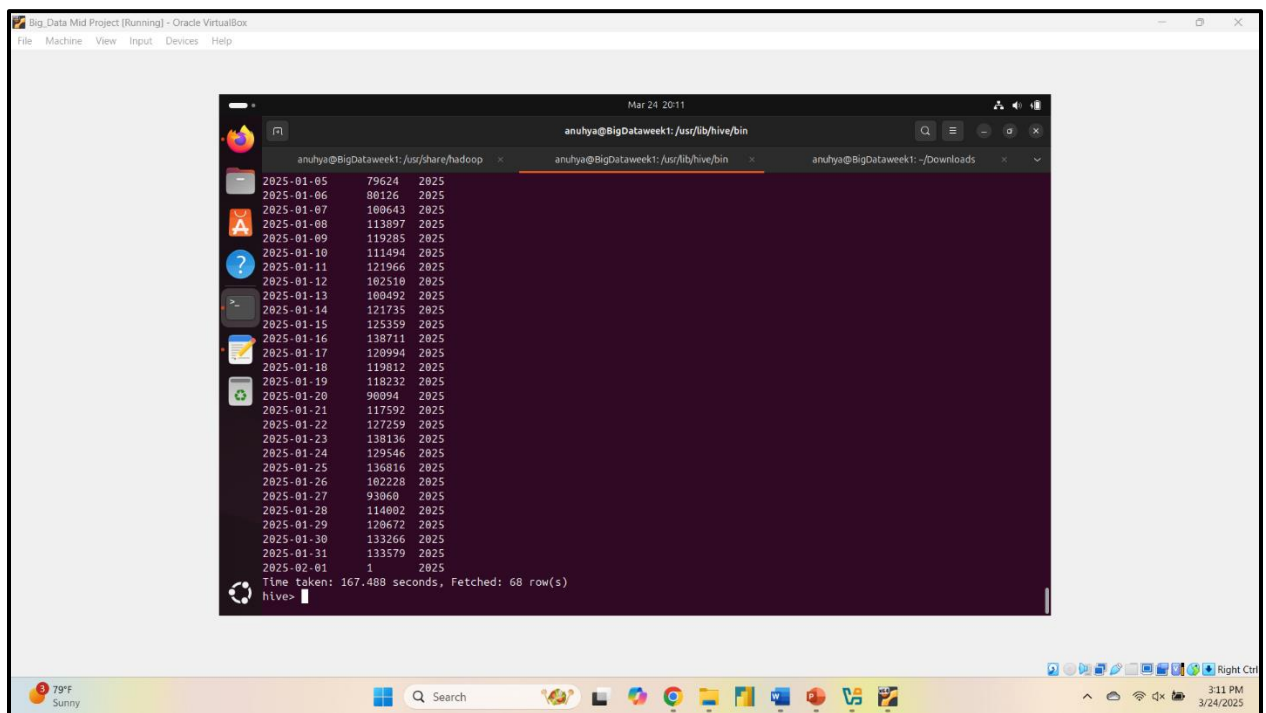
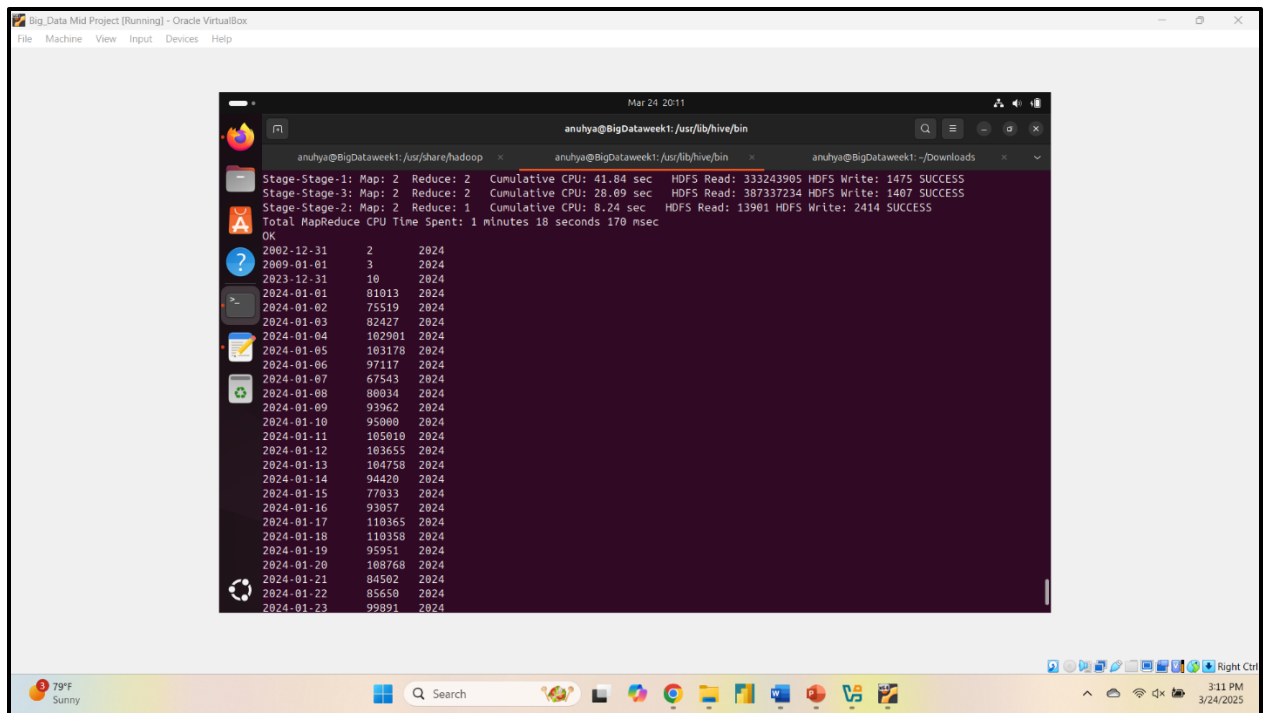
Total MapReduce CPU Time Spent: 42 seconds 290 msec
OK
2002-12-31      0.0      2024
2009-01-01    33.33333333333336      2024
2023-12-31    14.41      2024
2024-01-01    21.788021058348747      2024
2024-01-02    20.967073451714302      2024
2024-01-03    19.66402258968544      2024
2024-01-04    18.421313495495873      2024
2024-01-05    17.83245468995311      2024
2024-01-06    17.310473449550734      2024
2024-01-07    19.45297854699959      2024
2024-01-08    18.698506009945955      2024
2024-01-09    16.89826153125751      2024
2024-01-10    17.920641684210477      2024
2024-01-11    18.55664193886312      2024
2024-01-12    18.548339973952157      2024
2024-01-13    17.276198858321262      2024
2024-01-14    17.870277165854585      2024
2024-01-15    19.298005789726666      2024
2024-01-16    19.172022523829376      2024
2024-01-17    18.203946359806263      2024
2024-01-18    17.895973558781506      2024
2024-01-19    17.2211299517463      2024
```



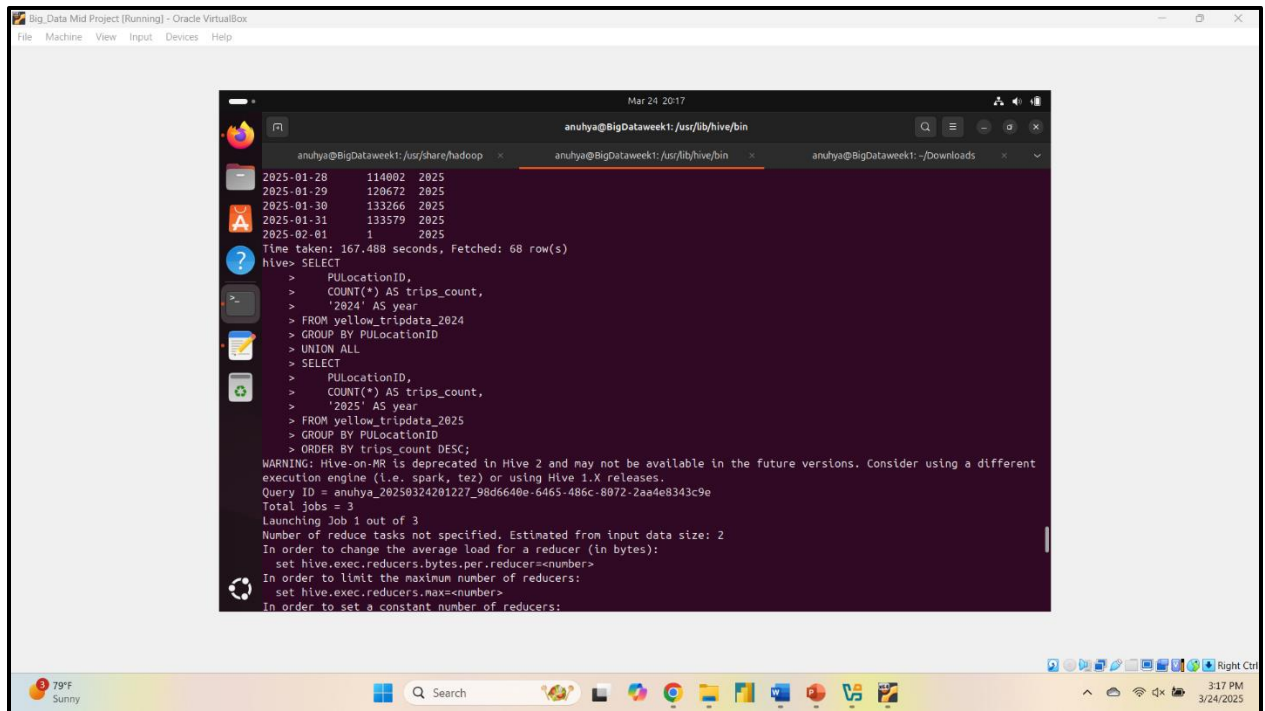
Anuhya Queries

Query 1



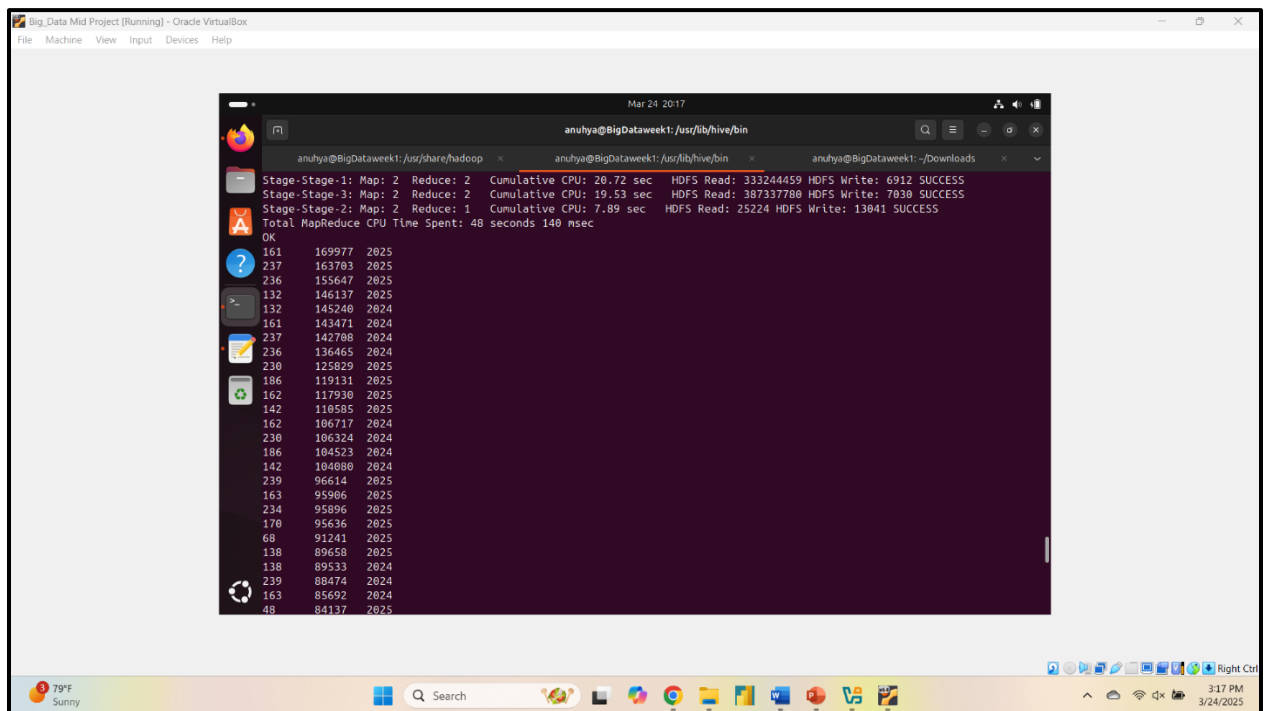


Query 2



```
anuhya@BigDataweek1: /usr/lib/hive/bin
anuhya@BigDataweek1: /usr/share/hadoop
anuhya@BigDataweek1: ~/Downloads

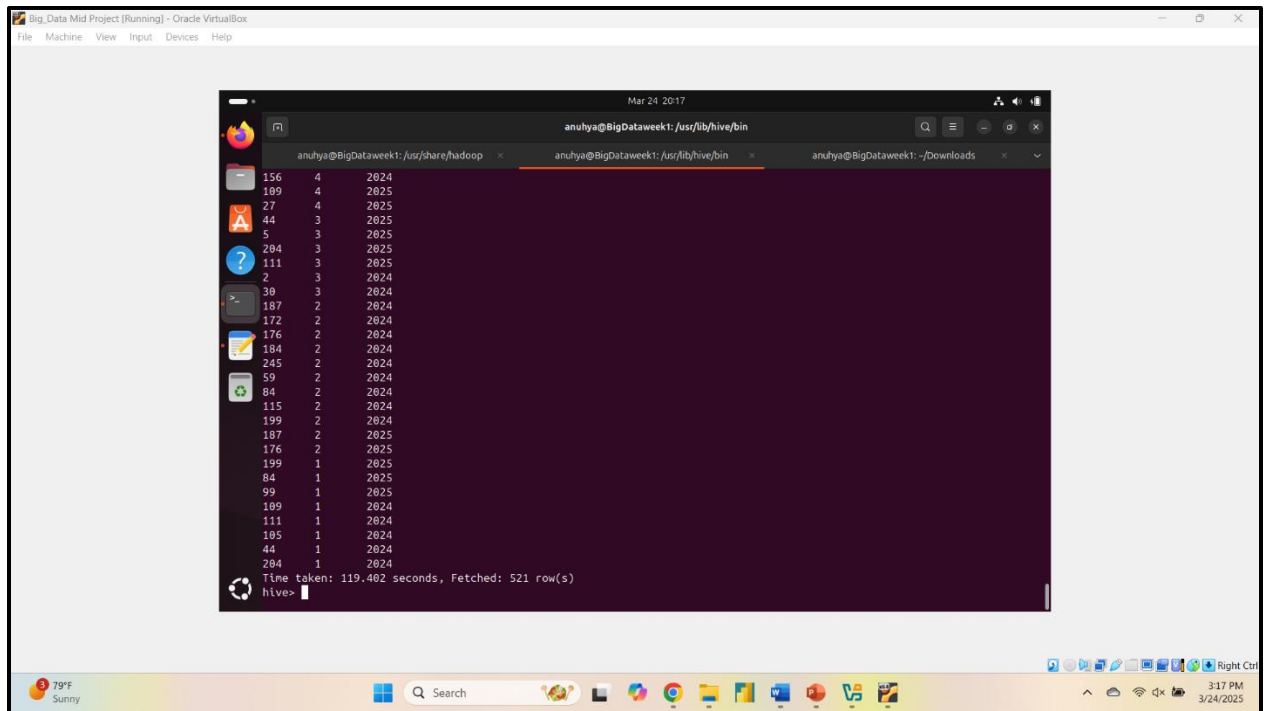
2025-01-28 114802 2025
2025-01-29 120672 2025
2025-01-30 133266 2025
2025-01-31 133579 2025
2025-02-01 1 2025
Time taken: 167.488 seconds, Fetched: 68 row(s)
hive> SELECT
>   PULocationID,
>   COUNT(*) AS trips_count,
>   '2024' AS year
> FROM yellow_tripdata_2024
> GROUP BY PULocationID
> UNION ALL
> SELECT
>   PULocationID,
>   COUNT(*) AS trips_count,
>   '2025' AS year
> FROM yellow_tripdata_2025
> GROUP BY PULocationID
> ORDER BY trips_count DESC;
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different
execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = anuhya_20250324201227_98d6640e-6465-486c-8072-2aa4e8343c9e
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks not specified, Estimated from input data size: 2
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
```



```
anuhya@BigDataweek1: /usr/lib/hive/bin
anuhya@BigDataweek1: /usr/share/hadoop
anuhya@BigDataweek1: ~/Downloads

Stage-Stage-1: Map: 2 Reduce: 2 Cumulative CPU: 20.72 sec HDFS Read: 333244459 HDFS Write: 6912 SUCCESS
Stage-Stage-3: Map: 2 Reduce: 2 Cumulative CPU: 19.53 sec HDFS Read: 387337780 HDFS Write: 7030 SUCCESS
Stage-Stage-2: Map: 2 Reduce: 1 Cumulative CPU: 7.89 sec HDFS Read: 25224 HDFS Write: 13041 SUCCESS
Total MapReduce CPU Time Spent: 48 seconds 140 msec
OK
161 169977 2025
237 163703 2025
236 155647 2025
132 146137 2025
132 145240 2024
161 143471 2024
237 142708 2024
236 136465 2024
230 125829 2025
186 119131 2025
162 117930 2025
142 110585 2025
162 106717 2024
230 106324 2024
186 104523 2024
142 104080 2024
239 96614 2025
163 95906 2025
234 95896 2025
170 95636 2025
68 91241 2025
138 89658 2025
138 89523 2024
239 88474 2024
163 85692 2024
48 84137 2025
```

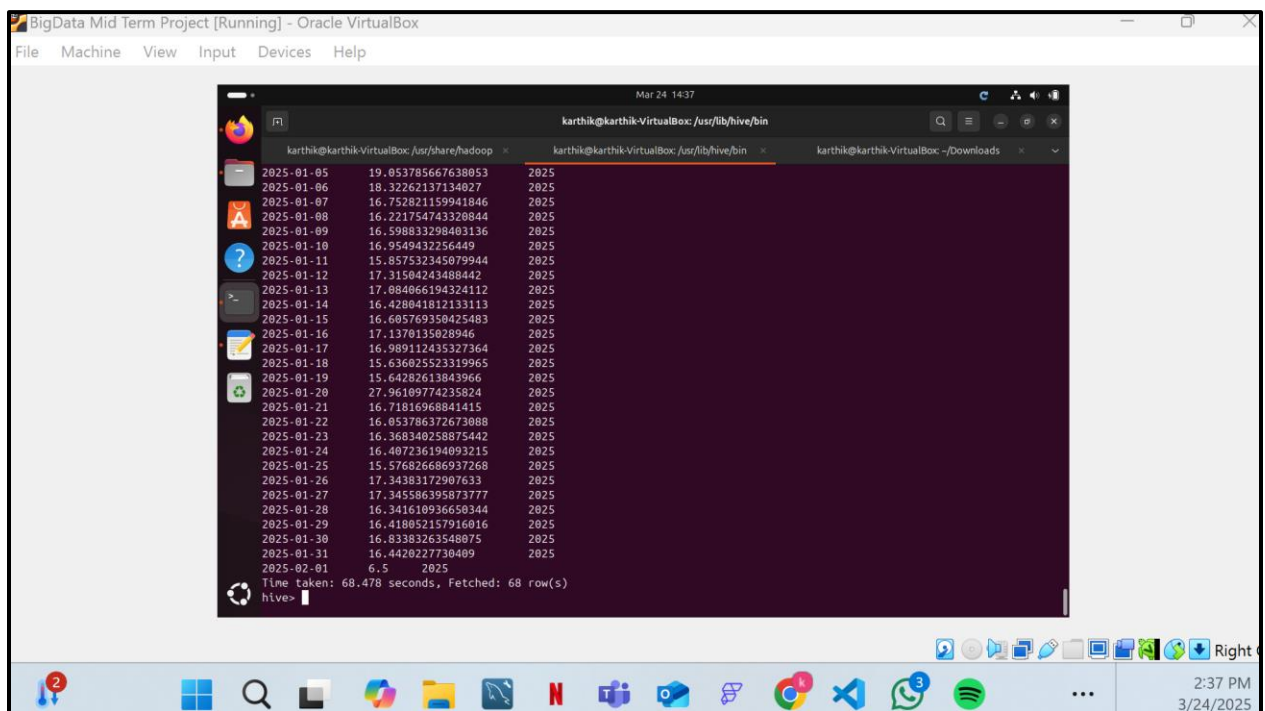

Query 3



```
anuhya@BigDataweek1: /usr/lib/hive/bin
anuhya@BigDataweek1: /usr/share/hadoop
anuhya@BigDataweek1: ~/Downloads

156 4 2024
109 4 2025
27 4 2025
44 3 2025
5 3 2025
284 3 2025
1111 3 2025
2 3 2024
38 3 2024
187 2 2024
172 2 2024
176 2 2024
184 2 2024
245 2 2024
59 2 2024
84 2 2024
115 2 2024
199 2 2024
187 2 2025
176 2 2025
199 1 2025
84 1 2025
99 1 2025
109 1 2024
111 1 2024
105 1 2024
44 1 2024
284 1 2024

Time taken: 119.402 seconds, Fetched: 521 row(s)
hive>
```



```
karthik@karthik-VirtualBox: /usr/lib/hive/bin
karthik@karthik-VirtualBox: /usr/share/hadoop
karthik@karthik-VirtualBox: ~/Downloads

2025-01-05 19.053785667638053 2025
2025-01-06 18.32262137134027 2025
2025-01-07 16.752821159941846 2025
2025-01-08 16.221754743320844 2025
2025-01-09 16.598833208403136 2025
2025-01-10 16.9549432256449 2025
2025-01-11 15.857532345079944 2025
2025-01-12 17.31504243488442 2025
2025-01-13 17.084066194324112 2025
2025-01-14 16.428041812133113 2025
2025-01-15 16.605769350425483 2025
2025-01-16 17.1370135028946 2025
2025-01-17 16.989112435327364 2025
2025-01-18 15.636025523319965 2025
2025-01-19 15.64282613843966 2025
2025-01-20 27.06109774235824 2025
2025-01-21 16.71816968841415 2025
2025-01-22 16.853786372673088 2025
2025-01-23 16.368340258875442 2025
2025-01-24 16.407236194093215 2025
2025-01-25 15.576826686937268 2025
2025-01-26 17.34383172907633 2025
2025-01-27 17.345586395873777 2025
2025-01-28 16.341610936650344 2025
2025-01-29 16.418052157916016 2025
2025-01-30 16.83383263548075 2025
2025-01-31 16.4420227730409 2025
2025-02-01 6.5 2025

Time taken: 68.478 seconds, Fetched: 68 row(s)
hive>
```